

FNCE30010

Algorithmic Trading

Semester 2, 2024

Project 2 (Backtesting)

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Selected Strategy

Strategy Description

Our strategy is based on CAPM Alpha Ranking applied to the Dow 30 stocks, as implemented on QuantConnect. This strategy leverages the Capital Asset Pricing Model (CAPM), a foundational stock pricing model developed by William F. Sharpe and Harry Markowitz. CAPM estimates an asset's expected return as a function of its risk relative to the market, represented by beta, and its excess return over the risk-free rate.

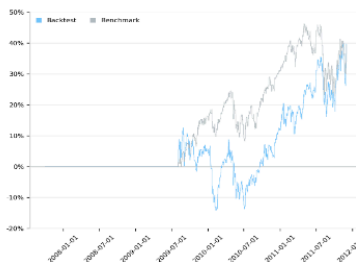
The strategy generates alpha, which represents the excess return of an asset relative to its expected CAPM return. Higher alpha values indicate a greater potential for outperforming the market. The underlying assumption in the strategy is that past high-alpha stocks will continue to outperform in the subsequent period. The strategy invests in two stocks with the highest alpha rankings from the previous month in equal proportions with leverage utilisation for enhanced exposure and expects them to outperform. These stocks are rebalanced monthly, with the DOW30 serving as the benchmark for performance comparison.

Strategy Results

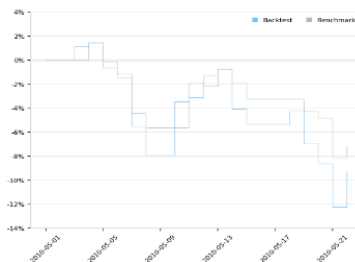
Cumulative Returns



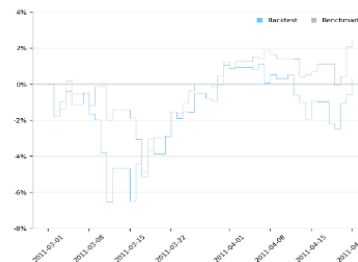
Global Financial Crisis 2007



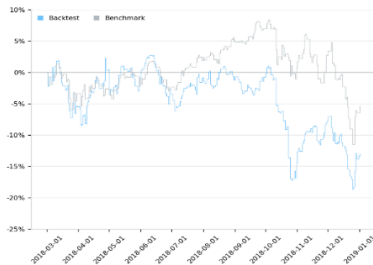
Flash Crash 2010



Fukushima Meltdown 2011



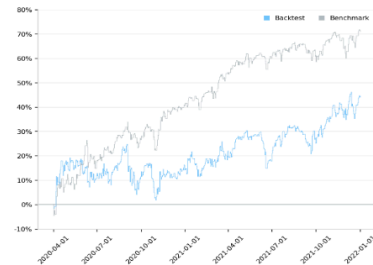
New Normal 2014-2019



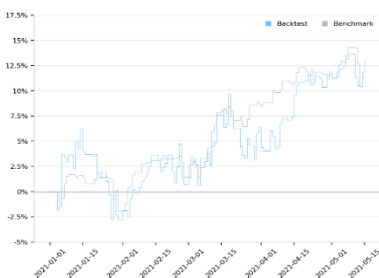
COVID-19 Pandemic 2020



Post-COVID Run-up 2020-2021



Meme Season 2021



Russia Invades Ukraine 2022-2023



AI Boom 2022-Present



Key Statistics

Runtime Days	2193	Drawdown	32.8%
Turnover	6%	Probabilistic SR	5%
CAGR	10.2%	Sharpe Ratio	0.3
Capacity (USD)	1.3M	Sortino Ratio	0.4
Trades per Day	0.2	Information Ratio	0.1

Monthly Returns

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2018												
2019												
2020												
2021												
2022												
2023												
2024												

The chosen strategy focusing on Dow 30 stocks achieved a CAGR of 10.2%, with a Sharpe ratio of 0.3, a drawdown of 32.8%, and a beta of 1.008 across 5 years. The strategy generally lagged behind the benchmark, with strong drawdowns in times of volatility, especially during the global financial crisis and post-COVID recovery period. This underperformance can be attributed to its exposure to high-risk assets and a lack of defensive positioning during market downturns. Additionally, the strategy's reliance on specific stock selections may have resulted in concentrated losses, further exacerbating the drawdowns. As a result, the inability to adapt to changing market conditions highlighted the importance of risk management and diversification in maintaining performance during turbulent times. The strategy only started to outperform in 2022 after the AI boom. This performance shift raises concerns about the strategy's predictive power, as the earlier negative alpha indicated a failure to generate excess returns relative to its risk. The post-2022 outperformance suggests that the strategy was able to leverage the momentum created by AI advancements, rather than relying on consistent alpha generation.

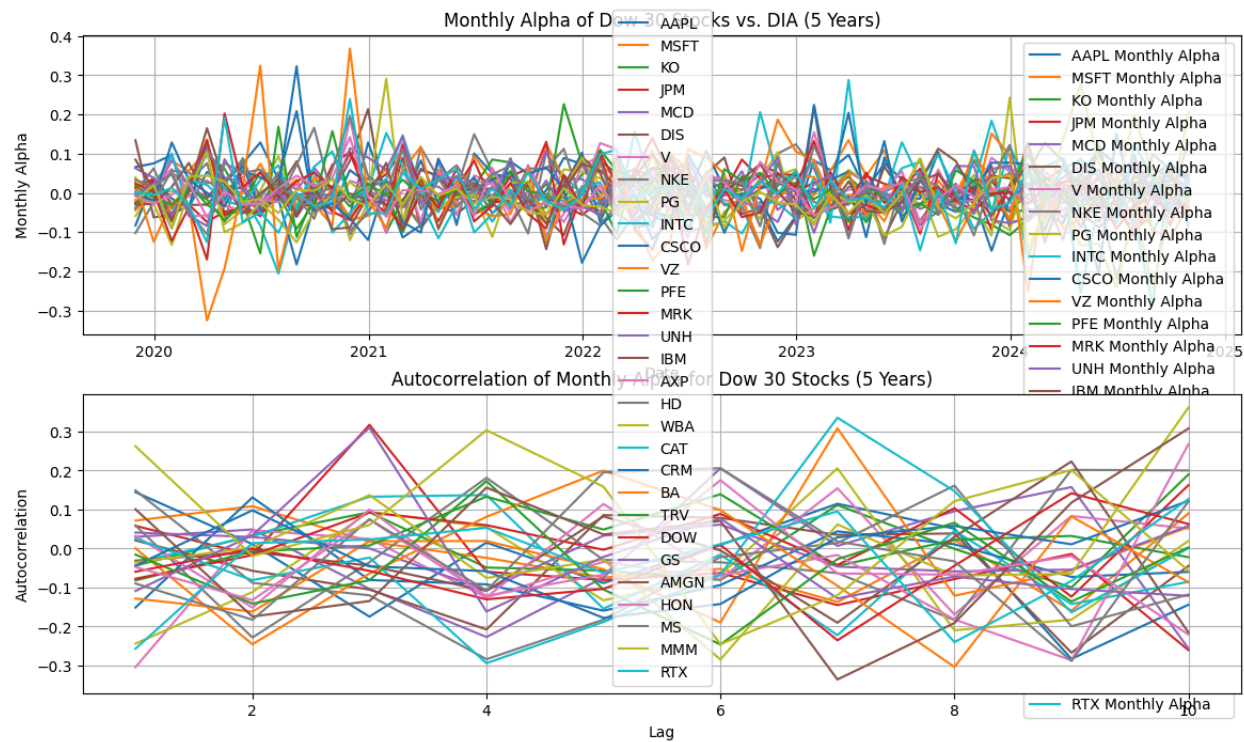
Motivation of Strategy

The price momentum strategy, as identified by Jegadeesh and Titman (1993), shows that stocks with strong past performance tend to outperform over short- to medium-term horizons. Similarly, past short-term alphas (6–12 months) can positively forecast future returns, while long-term alphas (6–120 months, lagged by 12 months) exhibit a negative correlation. Notably, the alpha momentum effect subsumes the price momentum effect, fully explaining the profits from price momentum, yet remaining unexplained by it (Zaremba, A., Umutlu, M., & Karathanasopoulos, A., 2019).

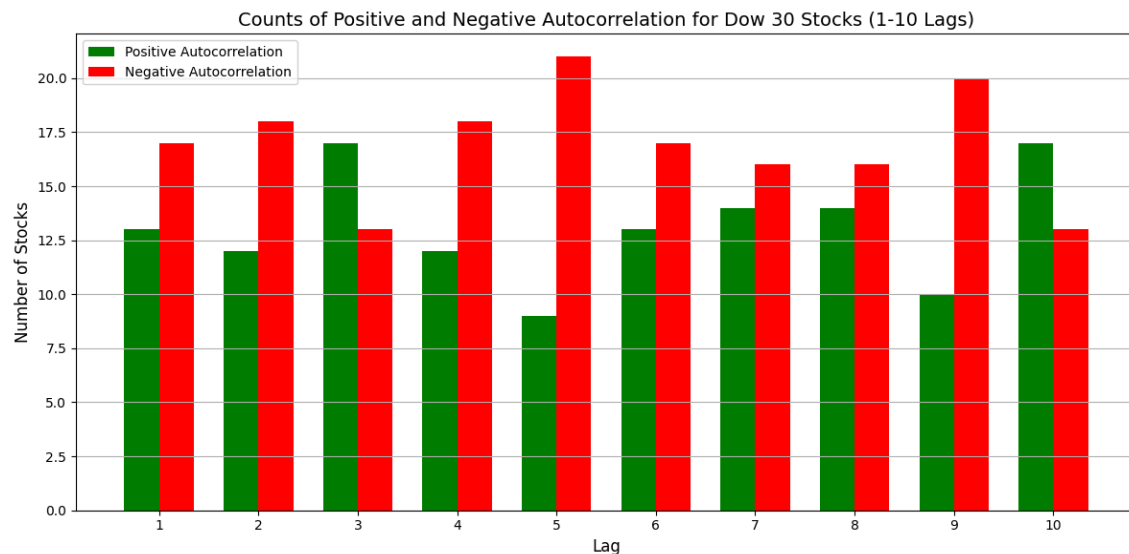
The alpha momentum strategy focuses on investing in assets with strong recent alpha, demonstrating superior returns compared to traditional price-based momentum strategies in both the U.S. and Europe. Zero-investment portfolios based on past alpha have yielded higher returns in the U.S. and displayed lower volatility across both regions (Hühn & Scholz, 2018).

Analysis of long-term performance reveals that alpha momentum does not reverse as sharply as price momentum. Price momentum is driven by overshooting due to momentum trading, while alpha momentum is linked to underreaction to firm-specific news. Both strategies are sensitive to market sentiment, showing stronger profits after optimistic periods.

Positive alphas may indicate potential arbitrage opportunities or dominated securities rather than simply reflecting abnormal expected returns. This aligns with Jarrow and Protter's findings, suggesting that while positive alpha hints at market inefficiency, such opportunities are rarer than abnormal returns. The infrequency of these arbitrage opportunities implies that positive alpha strategies may often arise from omitted variables rather than true arbitrage. Furthermore, Levy and Roll (2016) highlight that positive alpha can lead to incorrect adjustments, suggesting that assets with positive alpha should be underweighted and vice versa. This emphasises the need for careful analysis when interpreting positive alpha, as it may not represent consistent investment advantages in practice.



The charts illustrate the monthly alpha values and autocorrelation of Dow 30 stocks over a five-year period. The top chart displays fluctuating monthly alpha values, with stocks alternating between outperforming and underperforming the benchmark (DIA). Most alphas converge near zero over time, indicating that while some stocks show spikes in performance, consistent outperformance is rare. The bottom chart reveals low autocorrelation of monthly alpha values across different lags, suggesting limited predictive power for future performance. This indicates that recent alpha may not reliably predict future stock outperformance, as most Dow 30 stocks do not demonstrate strong persistence in relative returns.



We observed that negative autocorrelation predominantly influences stock performance over time, indicating a tendency for performance reversals. In contrast, positive autocorrelation, which signifies performance continuity, is less prevalent and appears more balanced with negative autocorrelation at lags 2 and 10. This pattern suggests mean-reversion behaviour in Dow 30 stocks, indicating that an alpha momentum strategy may be less effective.

However, several Dow 30 stocks, including Apple (AAPL) and Microsoft (MSFT), display consistent positive autocorrelation across multiple lags, demonstrating strong trend-following behaviour. Other stocks like Pfizer (PFE), Amgen (AMGN), and 3M (MMM) also exhibit persistent trends, suggesting these stocks may maintain momentum over time. Additionally, consumer brands such as McDonald's (MCD), Disney (DIS), and Visa (V) show similar trends, indicating resilience and potential for sustained movements. This trend extends to UnitedHealth (UNH) and Raytheon Technologies (RTX), making these stocks suitable candidates for alpha momentum-based strategies.

Weaknesses

Reliance on Short-Term Alpha

Alpha exhibits limited predictive power for most stocks, showing weak persistence in relative returns and raising doubts about its effectiveness as a performance predictor. Gallagher (2024) notes that traditional CAPM struggles to capture return variations, while nonparametric methods with variable factor loadings offer a more accurate reflection of evolving market risk factors. This unreliability in alpha poses risks in portfolio construction, as the strategy heavily depends on alpha derived from the CAPM model.

Additionally, prevalent negative autocorrelation suggests a mean-reversion bias, contradicting the notion that trends will persist, potentially leading to losses for investors expecting continued outperformance. The strategy's focus on the previous month's alpha may be skewed by short-term volatility and noise, leading to poor investment decisions that misinterpret performance signals. This reliance can prompt the algorithm to buy stocks based on temporary alpha spikes or drops, ultimately affecting overall strategy performance.

Over-Concentration

The strategy involves constructing a concentrated portfolio that lacks diversification across sectors or asset classes. This lack of diversification makes it vulnerable to specific negative developments and limits exposure to other market opportunities. Without sector diversification, the strategy may miss high-performing stocks and opportunities that could boost overall returns. Diversification not only helps manage risk but also enables the portfolio to adapt to varying market conditions, as different sectors respond differently to economic cycles.

Market Volatility Sensitivity

The current strategy's dependence on high alpha stocks heightens its sensitivity to market volatility. During periods of significant fluctuations, these stocks can behave unpredictably, leading to unexpected performance downturns. Factors like investor sentiment, macroeconomic changes, and geopolitical events can trigger rapid price shifts, affecting returns. Research indicates that alpha is sensitive to changes in portfolio composition; regular rebalancing can lead to significant and sometimes unpredictable changes in alpha (Levy & Roll, 2016).

Moreover, stocks that excel in stable conditions may not maintain their alpha during volatility. Focusing on recent alpha rankings might lead to heavy investments in these stocks just as they start to underperform. Consequently, this strategy could incur larger losses in downturns when high alpha stocks undergo significant price corrections. This sensitivity to market conditions suggests the need for a more diversified approach to mitigate potential losses during turbulent times.

Lack of Macro-Economic Context

The strategy's reliance on alpha rankings overlooks the broader macroeconomic factors that significantly influence stock performance. According to Savor & Wilson (2013), 60% of annual stock returns occur on macroeconomic news days. alpha-focused strategy like this does not account for broader economic trends like inflation, interest rates, or geopolitical risks, which could increase portfolio risk during economic instability. Additionally, by not accounting for macroeconomic factors, the strategy may miss opportunities to adjust its asset allocation in anticipation of broader market trends.

Overfitting and Seasonality Effects

A significant risk in the current strategy is overfitting, where the algorithm is overly adapted to historical data, capturing patterns that may not persist in real markets. This can lead to strong backtest performance but weak real-time results, as the model might respond to past anomalies instead of genuine trends.

Additionally, the strategy overlooks seasonal patterns affecting stock performance, such as consumer behaviour and corporate earnings cycles. Ignoring these cyclical factors can result in misinterpreting short-term alpha increases as signals of long-term outperformance, potentially leading to investments in stocks that are only temporarily high-performing. Consequently, this can increase risk and result in missed opportunities if the algorithm fails to anticipate cyclical downturns.

Increased Transaction Costs

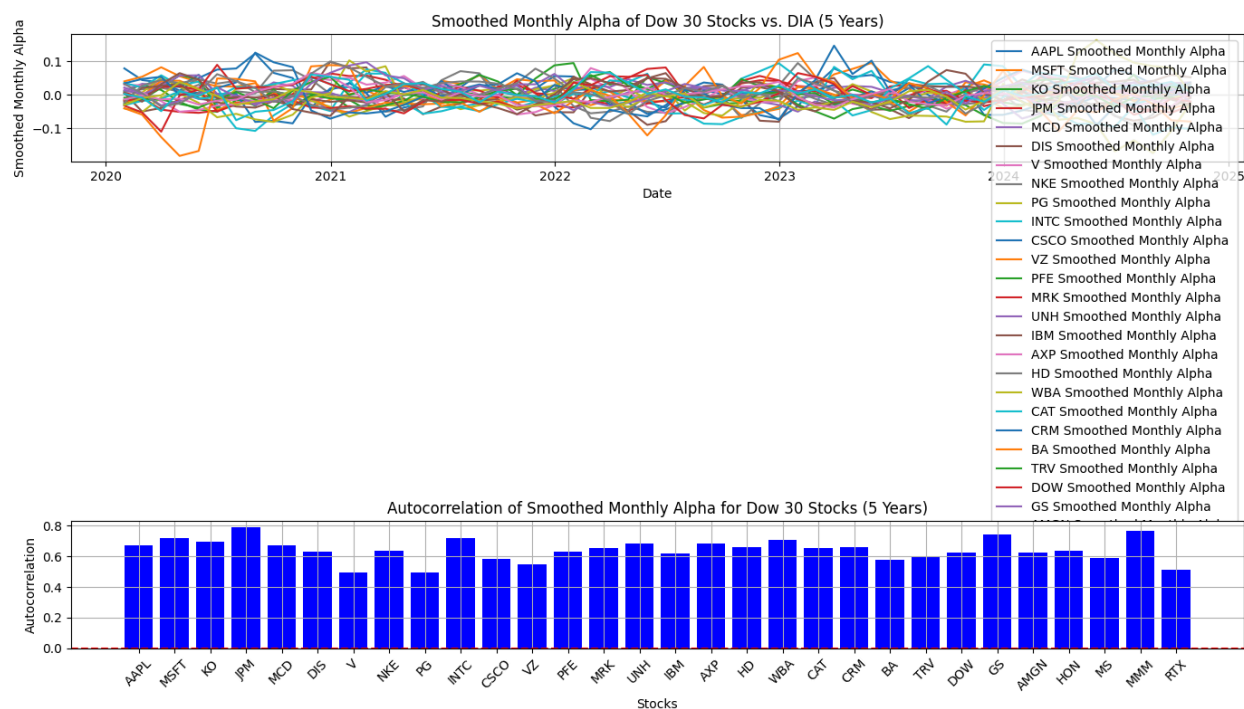
The strategy incurs higher transaction costs due to its monthly rebalancing, which necessitates frequent trades. This results in increased brokerage fees, slippage, and taxes, particularly in low-liquidity markets. For smaller portfolios, these costs can take up a significant portion of invested capital, eroding returns over time. Additionally, frequent trading can lead to slippage, further diminishing profitability.

Improvements & Evaluation

Alpha Smoothing

Alpha smoothing is essential for reducing the strategy's dependence on short-term alpha. By averaging alpha over a longer period, the algorithm becomes less affected by short-term market fluctuations and noise, leading to more stable performance trends instead of being swayed by temporary price movements. This approach enhances the strategy's reliability by focusing on enduring performance signals rather than transient spikes.

Additionally, alpha smoothing mitigates the risk of overfitting, where a model too closely aligns with historical data may misidentify non-recurring patterns. By emphasising broader performance trends, the algorithm can better generalise to new data and future market conditions, resulting in a more robust investment model.



All calculated autocorrelation values for the smoothed monthly alpha of the Dow 30 stocks were found to be significant, exceeding 0.5. This strong positive autocorrelation reinforces the notion that smoothed alpha is a reliable predictor of future stock performance, instead of relying on a single month's alpha.

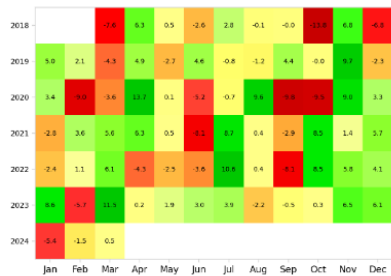
Hence, to address the assumption of reliance on short-term alpha and to mitigate the risk of overfitting, we will implement alpha smoothing in our strategy. Rather than relying solely on a

single month’s data to rank stocks, we will extend the evaluation period to 3 months to consider multiple alpha values and calculate their average. This enhanced alpha-ranking method enables the algorithm to incorporate a broader range of information, filtering out stocks that may have performed well due to temporary market overreactions.

Key Statistics

Runtime Days	2193	Drawdown	32.8%
Turnover	6%	Probabilistic SR	5%
CAGR	10.2%	Sharpe Ratio	0.3
Capacity (USD)	1.3M	Sortino Ratio	0.4
Trades per Day	0.2	Information Ratio	0.1

Monthly Returns

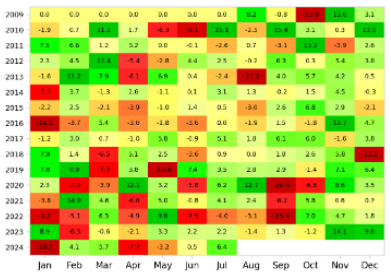


Before Alpha Smoothing

Key Statistics

Runtime Days	5682	Drawdown	38.6%
Turnover	4%	Probabilistic SR	2%
CAGR	12.9%	Sharpe Ratio	0.5
Capacity (USD)	700K	Sortino Ratio	0.5
Trades per Day	0.2	Information Ratio	0.1

Monthly Returns



After Alpha Smoothing

After incorporating alpha smoothing, we found that the CAGR increased to 12.9% and the Sharpe ratio rose to 0.5 for DOW30 stocks. However, this led to a significant increase in drawdown to 38.6%. This increase may be attributed to the smoothing process delaying the recognition of declining performance in individual stocks. Consequently, stocks experiencing short-term underperformance but with historically positive returns or alpha might remain in the portfolio longer than warranted (Huang, J., Liechty, J., & Rossi, M., 2018). In volatile market conditions, correlations between stocks can also result in simultaneous declines across sectors, exacerbating drawdown effects. Therefore, while alpha smoothing seeks to stabilise stock assessments, it may inadvertently lead to larger drawdowns by failing to react promptly to negative trends.

Risk Parity Model - Maximum Sharpe Ratio Optimizer

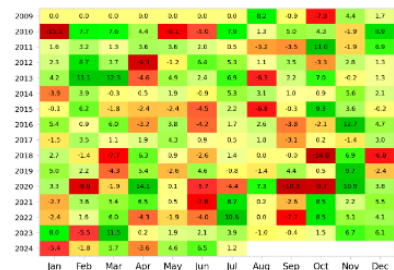
Implementing a Max Sharpe Ratio approach within our CAPM Alpha Ranking strategy significantly enhances risk-adjusted performance by optimising stock weights based on risk and expected return. This balances potential returns with volatility, resulting in a more stable portfolio. The risk parity method further improves the Sharpe ratio by adjusting asset allocations to maintain consistent risk levels, enhancing resilience during market downturns (Feng & Palomar, 2015, 2016).

By integrating risk parity with Max Sharpe Ratio optimization, we achieve equitable risk distribution, reducing significant loss potential from over-concentration in any single stock or sector. This combined approach mitigates risks and enhances diversification by tailoring stock weights to risk-adjusted returns, decreasing reliance on high-alpha stocks. Implementing this strategy fosters a well-diversified portfolio that effectively manages risks while optimising performance.

Key Statistics

Runtime Days	5678	Drawdown	32.5%
Turnover	6%	Probabilistic SR	5%
CAGR	15.8%	Sharpe Ratio	0.6
Capacity (USD)	2.6M	Sortino Ratio	0.7
Trades per Day	0.2	Information Ratio	0.3

Monthly Returns

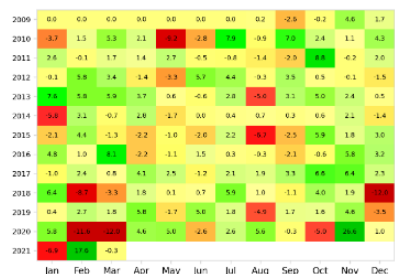


Risk-Parity Model - Max Sharpe Ratio Optimiser

Key Statistics

Runtime Days	4445	Drawdown	39.0%
Turnover	6%	Probabilistic SR	7%
CAGR	12.8%	Sharpe Ratio	0.6
Capacity (USD)	1.6M	Sortino Ratio	0.6
Trades per Day	2.4	Information Ratio	0.0

Monthly Returns

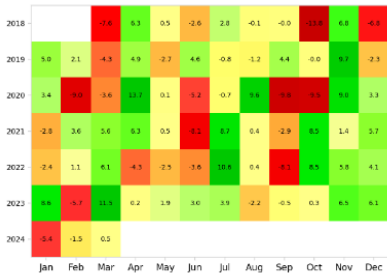


Mean-Variance Optimization Model - Max Sharpe Ratio Optimiser

Key Statistics

Runtime Days	2193	Drawdown	32.8%
Turnover	6%	Probabilistic SR	5%
CAGR	10.2%	Sharpe Ratio	0.3
Capacity (USD)	1.3M	Sortino Ratio	0.4
Trades per Day	0.2	Information Ratio	0.1

Monthly Returns



Equal Weighted Portfolio

Research indicates that the risk-parity (RP) model, especially when employing the Sharpe strategy, delivered significantly higher average daily returns and enhanced diversification compared to mean-variance optimization (MVO) portfolios, which tended to be more sector-focused. All portfolios showed lower overall risk than the market portfolio, with the RP model's correlation strategy particularly excelling in risk-adjusted returns. The RP strategy notably performed well during turbulent market conditions, such as the COVID-19 pandemic, avoiding negative returns and showcasing effective risk management (AlHalaseh & Al Shawawreh, 2024).

To further validate this theory, we also implemented the mean-variance model. Our backtesting reaffirmed that the risk-parity model and the maximisation of the Sharpe ratio achieved superior performance. The risk-parity model exhibited the lowest drawdown and achieved the highest compound annual growth rate (CAGR), Sharpe ratio, and information ratio.

Beta

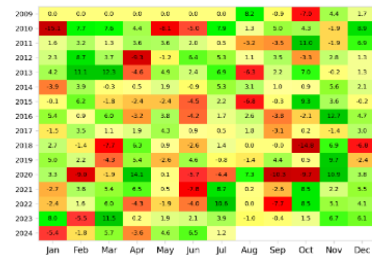
Integrating beta analysis into our CAPM Alpha Ranking strategy by selecting stocks with a beta close to one fosters a balanced risk-return profile, as these stocks generally move in line with the market, offering moderate risk and volatility (Sharpe, 1964). This approach enhances portfolio stability during market fluctuations, reducing the likelihood of dramatic price swings and enabling resilience in uncertain conditions.

By adopting this beta-focused strategy, we aim to construct a robust investment framework that effectively seeks alpha-driven opportunities while managing risk. This balanced approach promotes stable performance relative to the broader market, diminishes the risk of significant drawdowns, and facilitates more predictable returns aligned with market trends.

Key Statistics

Runtime Days	5678	Drawdown	32.5%
Turnover	6%	Probabilistic SR	5%
CAGR	15.8%	Sharpe Ratio	0.6
Capacity (USD)	2.6M	Sortino Ratio	0.7
Trades per Day	0.2	Information Ratio	0.3

Monthly Returns

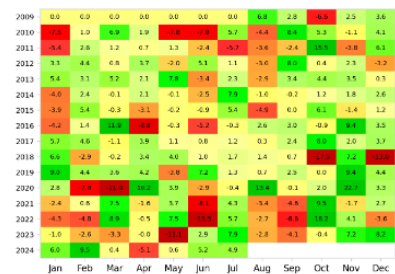


Before incorporating Beta

Key Statistics

Runtime Days	5682	Drawdown	35.9%
Turnover	6%	Probabilistic SR	4%
CAGR	13.5%	Sharpe Ratio	0.6
Capacity (USD)	4.9M	Sortino Ratio	0.6
Trades per Day	0.2	Information Ratio	0.1

Monthly Returns



After incorporating Beta

Incorporating stocks with a beta around 1 typically aligned with market movements; however, this approach has resulted in a significant drawdown of 35.9% and a lower compound annual growth rate (CAGR), along with a diminished Information Ratio. Such performance issues are linked to increased market volatility, which impacts all stocks similarly, causing larger drawdowns during downturns. While these stocks seem safer, they often fail to capitalise on the higher returns of higher-beta stocks that outperform in bull markets. Consequently, this strategy may inadvertently include lower-alpha stocks, as research indicates that higher beta is associated with declining alpha (Frazzini & Pedersen, 2014). Thus, the constraint of maintaining a beta around 1 could lead to a less favourable risk-return profile compared to more dynamic options. Hence, we would not incorporate this strategy.

Diversification

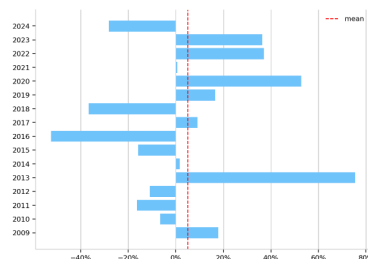
Enhancing diversification is essential for managing market volatility and concentration risks in our trading strategy. Spreading investments across various sectors and asset classes can cushion adverse price movements. A study indicates that portfolio risk decreases with an increasing number of stocks, stabilising after about twelve, which minimises diversification risk (Li, 2022). Studies demonstrate that there is significantly greater benefit in diversifying equity portfolios across sectors than within sectors as well (James, N., Menzies, M., & Gottwald, G. A., 2022).

To mitigate over-concentration, we'll include 10 carefully selected stocks, ensuring no sector exceeds 40%. This balanced approach leverages diverse market segments while shifting our benchmark from the DOW30 to the S&P 500 for greater diversification.

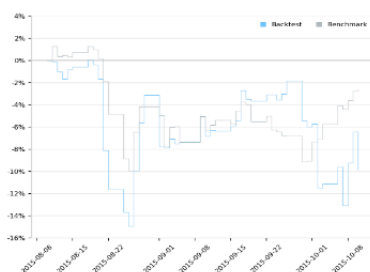
Key Statistics

Runtime Days	5682	Drawdown	80.1%
Turnover	6%	Probabilistic SR	0%
CAGR	0.0%	Sharpe Ratio	0.1
Capacity (USD)	830K	Sortino Ratio	0.1
Trades per Day	0.4	Information Ratio	-0.2

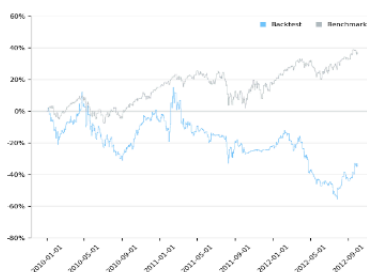
Annual Returns



Market Sell-Off 2015



Recovery 2010-2012



New Normal 2014-2019



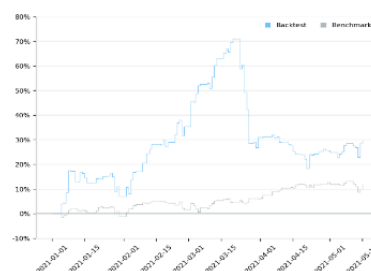
COVID-19 Pandemic 2020



Post-COVID Run-up 2020-2021



Meme Season 2021

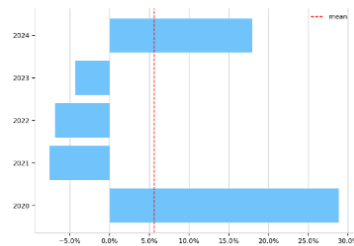


Portfolio of 2 S&P stocks

Key Statistics

Runtime Days	1666	Drawdown	36.2%
Turnover	11%	Probabilistic SR	3%
CAGR	4.9%	Sharpe Ratio	0.2
Capacity (USD)	850K	Sortino Ratio	0.2
Trades per Day	2.3	Information Ratio	-0.3

Annual Returns



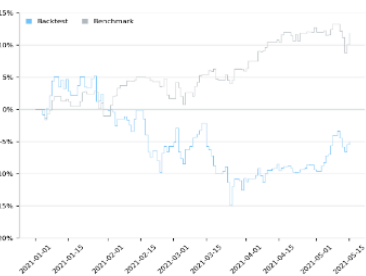
COVID-19 Pandemic 2020



Post-COVID Run-up 2020-2021



Meme Season 2021



Final Results of Improved Strategy

Firstly, we like to note that after changing the benchmark and stock selection to S&P 500, the strategy resulted in a net loss with a significant drawdown of 80.1%. This differs from using Dow 30, possibly due to the Dow 30's concentrated nature, allowing high alpha stocks to align closely with market trends, enhancing their performance. Conversely, the broader S&P 500, which includes 500 stocks, may see underperformance due to volatility or weaker correlations among selected high alpha stocks. The portfolio only outperformed the benchmark post Covid, possibly due to the tech sector dominance. Overall, this could imply a lack of effectiveness of strategy in the volatile S&P 500 index.

After incorporating diversification, the algorithm faced margin calls due to high portfolio volatility. Selecting 10 stocks increased exposure, as broader diversification made the portfolio more sensitive to market swings. During volatile periods, stocks can become more correlated, even across different sectors, leading to compounded losses and margin calls when equity fell below maintenance levels (Gao & Mei, 2019).

Analysing a shorter duration from 2020 to 2024 during the AI boom revealed significantly lower drawdowns. However, the diversified portfolio underperformed compared to a concentrated strategy with just two stocks. This supports findings that concentrated portfolios often yield superior returns and are biased towards positive momentum stocks, making them sensitive to market sentiment. While these portfolios can excel in bullish markets, they are more susceptible to larger losses during downturns (Yeung et al., 2012).

Limitations & Future Extension

Overall, while our implemented improvements have boosted risk-adjusted returns, a significant drawdown remains, particularly in regards to margin calls. This highlights the need for further efforts to reduce portfolio volatility and minimise exposure to sudden market swings that can trigger margin calls. The existing weaknesses, particularly regarding a lack of macroeconomic context, the impact of transaction costs and seasonality effects still need to be addressed. The strategy is also more effective in stable stocks, like in the Dow 30. Finally, past performance is no guarantee of future results.

To enhance our strategy, we should integrate macroeconomic indicators to adjust our focus toward more defensive stocks during volatile periods while also accounting for seasonality to avoid investing in stocks that may be temporarily outperforming due to short-term anomalies. We can also utilise fundamental analysis to select high-quality stocks rather than relying on sentiment-driven stocks. It is also a possibility to consider those stocks with positive alpha autocorrelation. Research shows that stop-loss policies can improve returns and reduce risk, particularly in momentum and regime-switching environments, often outperforming traditional buy-and-hold strategies (Kaminski & Lo, 2014; Chan et al., 2008). To reduce volatility, integrating stop-loss strategies into our CAPM Alpha Ranking framework can enhance risk management by establishing predetermined price thresholds for automatic stock sales, which limit losses and protect capital from market downturns.

References

- AlHalaseh, R. H., & Al Shawawreh, F. K. (2024). Enhancing portfolio optimization: A comparative analysis of the mean-variance Markowitz model and risk-parity contribution strategies. *Corporate & Business Strategy Review*, 5(3), 124–136. <https://doi.org/10.22495/cbsrv5i3art12>
- Asness, C. S., Frazzini, A., & Pedersen, L. H. (2019). Quality minus junk. *Review of Accounting Studies*, 24(1), 34–112. <https://doi.org/10.1007/s11142-018-9470-2>
- Danny Yeung & Paolo Pellizzari & Ron Bird & Sazali Abidin, 2012. "Diversification Versus Concentration and the Winner Is?," Working Paper Series 18, The Paul Woolley Centre for Capital Market Dysfunctionality, University of Technology, Sydney.
- Fama, E., & French, K. (2004). The Capital Asset Pricing Model: Theory and Evidence. *Journal of Economic Perspective*, 18(3), p25-46.
- Feng, Y., & Palomar, D. (2015). SCRIP: Successive convex optimization methods for risk parity portfolio design. *IEEE Transactions on Signal Processing*, 63(19), 5285–5300. <https://doi.org/10.1109/TSP.2015.2452219>
- Feng, Y., & Palomar, D. P. (2016). Portfolio optimization with asset selection and risk parity control. In 2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 6585–6589). Institute of Electrical and Electronics Engineers (IEEE). <https://doi.org/10.1109/ICASSP.2016.7472946>
- Frazzini, A., & Pedersen, L. H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1), 1–25. <https://doi.org/10.1016/j.jfineco.2013.10.005>
- Gallagher, M. (2024). MULTIFACTOR MODELS IN FINANCE WITH NONPARAMETRIC METHODS. *Journal of Business & Behavioral Science*, 36(1), p4-22.
- Gao & Mei (2019). The correlation structure in the international stock markets during global financial crisis. *Physica A*, 534, 122056-. <https://doi.org/10.1016/j.physa.2019.122056>
- Grinold, R. C. and Kahn, R.N. (2000) *Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk*. McGraw-Hill, New York.
- Huang, J., Liechty, J., & Rossi, M. (2018). Return smoothing and its implications for performance analysis of hedge funds. *The Journal of Finance and Data Science*, 4(4), 203–222. <https://doi.org/10.1016/j.jfds.2018.05.002>
- Hühn, H. L., & Scholz, H. (2018). Alpha momentum and price momentum. *International Journal of Financial Studies*, 6(2), 1–28. <https://doi.org/10.3390/ijfs6020049>

James, N., Menzies, M., & Gottwald, G. A. (2022). On financial market correlation structures and diversification benefits across and within equity sectors. *Physica A*, 604, 127682-. <https://doi.org/10.1016/j.physa.2022.127682>

Jarrow, R., & Protter, P. (2016). Positive alphas and a generalized multiple-factor asset pricing model. *Mathematics and Financial Economics*, 10(1), 29-48. [doi:https://doi.org/10.1007/s11579-015-0149-1](https://doi.org/10.1007/s11579-015-0149-1)

Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65-91.

Kaminski, K. M., & Lo, A. W. (2014). When do stop-loss rules stop losses? *Journal of Financial Markets* (Amsterdam, Netherlands), 18, 234–254. <https://doi.org/10.1016/j.finmar.2013.07.001>

Levy, M., & Roll, R. (2016). Seeking Alpha? It's a Bad Guideline for Portfolio Optimization. *Journal of Portfolio Management*, 42(5), p107-112.

Li, C. (2022). Empirical study on portfolio size and risk diversification. *Advances in Economics, Business and Management Research/Advances in Economics, Business and Management Research*. <https://doi.org/10.2991/aebmr.k.220307.479>

Savor, P., & Wilson, M. (2013). How Much Do Investors Care About Macroeconomic Risk? Evidence from scheduled Economic Announcement. *Journal of Financial & Quantitative Analysis*, 48(2), p343-375.

Sharpe, W. F. (1964). Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk. *The Journal of Finance*, 19(3), 425–442. <https://doi.org/10.2307/2977928>

Zaremba, A., Umutlu, M., & Karathanasopoulos, A. (2019). Alpha momentum and alpha reversal in country and industry equity indexes. *Journal of Empirical Finance*, 53, 144–161. <https://doi.org/10.1016/j.jempfin.2019.07.003>

Appendix - Code

```
import numpy as np

from AlgorithmImports import *

from universe import QC500UniverseSelectionModel


class CAPMAAlphaRankingModel(AlphaModel):

    def __init__(self, benchmark, lookback, alpha_smoothing_period=3):

        self.benchmark = benchmark

        self.lookback = lookback

        self.alpha_smoothing_period = alpha_smoothing_period

        self.symbols = []

        self.month = -1

        self.historical_alphas = {}

    def update(self, algorithm, data):

        if algorithm.time.month == self.month:

            return []

        self.month = algorithm.time.month

        returns = algorithm.history(

            self.symbols + [self.benchmark],

            self.lookback,

            Resolution.DAILY).close.unstack(level=0).pct_change().iloc[1:].fillna(0)

        return [

            Insight.price(symbol, Expiry.END_OF_MONTH, InsightDirection.UP)

            for symbol in self.select_symbols(algorithm, returns)

        ]

    def select_symbols(self, algorithm, all_returns):
```

```

alphas = {}

sector_counts = {}

benchmark = all_returns[self.benchmark]

for symbol in self.symbols:

    if str(symbol) not in all_returns.columns:

        algorithm.log(f'{symbol} not found in the historical data request')

        continue

    returns = all_returns[symbol]

    bla = np.vstack([benchmark, np.ones(len(returns))]).T

    result = np.linalg.lstsq(bla, returns, rcond=None)

    alpha = result[0][1]

    alphas[symbol] = alpha

    if symbol not in self.historical_alphas:

        self.historical_alphas[symbol] = []

    self.historical_alphas[symbol].append(alpha)

    if len(self.historical_alphas[symbol]) > self.alpha_smoothing_period:

        self.historical_alphas[symbol] =
self.historical_alphas[symbol][-self.alpha_smoothing_period:]

        smoothed_alpha = np.mean(self.historical_alphas[symbol])

        alphas[symbol] = smoothed_alpha

    selected = sorted(alphas.items(), key=lambda x: x[1], reverse=True)

    final_selected = []

    sector_selected_counts = {}

    for symbol, alpha in selected:

        security = algorithm.securities[symbol]

        sector = None

        if security.fundamentals and security.fundamentals.has_fundamental_data:

```

```

        sector = security.fundamentals.asset_classification.morningstar_sector_code

    if sector:

        if sector not in sector_selected_counts:

            sector_selected_counts[sector] = 0

        if sector_selected_counts[sector] < 4:

            final_selected.append(symbol)

            sector_selected_counts[sector] += 1

    if len(final_selected) >= 10:

        break

    return final_selected

def on_securities_changed(self, algorithm, changes):

    for added in changes.added_securities:

        self.symbols.append(added.symbol)

    for removed in changes.removed_securities:

        symbol = removed.symbol

        if symbol in self.symbols:

            self.symbols.remove(symbol)

        if symbol in self.historical_alphas:

            del self.historical_alphas[symbol]

from AlgorithmImports import *

from universe import QC500UniverseSelectionModel

class QC500UniverseSelectionModel(ETFConstituentsUniverseSelectionModel):

    def __init__(self, benchmark, universe_settings: UniverseSettings = None) -> None:

        super().__init__(benchmark, universe_settings, self.etf_constituents_filter)

```

```

def etf_constituents_filter(self, constituents):

    return [c.symbol for c in constituents]


#region imports

from AlgorithmImports import *

from universe import QC500UniverseSelectionModel

from alpha import CAPMAAlphaRankingModel

#endregion


class CapmAlphaRankingAlgorithm(QCAlgorithm):

    def initialize(self):

        self.set_start_date(2020, 1, 1)

        self.set_end_date(2024, 10, 1)

        self.set_cash(100000)      # Set Strategy Cash

        # Set number days to trace back

        lookback = self.get_parameter("lookback", 21)

        self.set_brokerage_model(BrokerageName.INTERACTIVE_BROKERS_BROKERAGE,
AccountType.MARGIN)

        spy = self.add_equity("SPY",

            resolution = self.universe_settings.resolution,

            data_normalization_mode = self.universe_settings.data_normalization_mode).symbol

        self.set_benchmark(spy)

        self.set_universe_selection(QC500UniverseSelectionModel(spy, self.universe_settings))

        self.add_alpha(CAPMAAlphaRankingModel(spy, lookback))

        self.set_portfolio_construction(RiskParityPortfolioConstructionModel(Expiry.END_OF_MONTH,
portfolio_bias= PortfolioBias.LONG, optimizer=MaximumSharpeRatioPortfolioOptimizer()))

```